

CO1 - Assessment Tool

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DSA 4605 - slot-C
Data handling and
visualization

- ① Explain the concept of mapping data onto aesthetics in data visualization.

Introduction:

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Data visualization is the process of representing data using charts, graphs, and maps. It helps people understand information quickly.

Types of Data:

- ① Categorical Data: It represents different groups or categories.

Example:

- Gender (Male, Female)
- Department

Visual Aesthetics: color, shape

- ② Ordinal Data:

It has a meaningful order.

Examples:

- Poor, Average, Good
- Small, medium, large

Visual Aesthetics:

- ordered colors, size

- ③ Quantitative Data:

- It contains numerical values.

Example:

- Age, salary, marks

Visual Aesthetics:

Position: shows the location of data on x and y axis.

Size and shape: represent magnitude and distinguishes different categories.

Advantages:

- easy to understand
- shows trends and makes comparison easier.
- Improves decision-making.

Conclusion:

mapping data onto aesthetic converts raw data into meaningful visual information.

② Describe the role of scales in data visualization.

A scale converts data values into visual position sizes or color in a graph. It ensures data accuracy.

Types of scales:

Linear scale:

Equal intervals represent equal values.

Example: 0, 10, 20, 30

used in: Bar charts, Line charts.

Logarithmic scale:

Each interval increase by multiplication..

Example: 1, 10, 100, 1000.

used in: Population growth, scientific data.

ordinal scale:

Represents ordered categories.

Example: Low, medium, high

different colors or shades represent different values

Example: Light Blue = Low rainfall, Dark Blue = High rain.

impact of incorrect scaling:

- mislead viewers and lead to incorrect conclusion.
- Exaggerate or hide trends.

Example: starting the y-axis from 90 instead of 0 makes small increase look much larger.

conclusion:

Scales are important because they ensure data is represented fairly, accurately, and clearly.

Discuss the importance of coordinate systems and axes in data visualization.

A coordinate system helps locate data points on a graph, while axes provide reference lines of measuring.

- x-axis: Horizontal

- y-axis: vertical

Cartesian coordinate system:

uses x-axis and y-axis to represent data.

→ Bar chart

→ Line chart

→ Scatter plot.

Advantage:-

- Easy to understand
- Accurate representation.
- Suitable for most data.

Polar coordinate system:-

uses radius and angles to represent data

Example:-

- Pie chart
- Radar chart

Advantages:-

- Best for circular data.
- Good for showing proportions.

Comparison:-

Cartesian	Polar
x and y axis's	Radius and Angle
Rectangular	circular
line & Bar charts	Pie & Radar charts

Conclusion:-

coordinate system and axes help organize data, compare values and display information clearly. Choosing the correct coordinate system improves visualization.

Explain the significance of nonlinear axes in data visualization.

A nonlinear axis does not increase by equal intervals. It is useful when data values vary.

Types of nonlinear axes:-

Logarithmic scale:-

values increase by multiplication.

Example:- 10, 100, 1000.

Other nonlinear scales:-

→ Square root, Exponential scale.

Advantages:-

- Displays large and small values together
- Reduce overcrowding.
- makes trends easier to identify.

Applications:-

- population growth.
- Earthquake intensity
- Stock market.
- Scientific research.

Conclusion:-

nonlinear axes are useful for wide-ranging datasets and help reveal patterns that may be visible with linear scales.

⑤ describe the role of color in data visualization.
- color is an important element in data visualization. It makes charts attractive, improve understanding, and helps users identify patterns and categories.

uses of color:-

Distinguishing categories:-

Different colors represent different groups.

Representing Data values:-

Different shades indicate different values.

Highlighting Important Information:-

Bright colors attract attention.

examples:-

Red = loss, Green = Profit.

Principles of choosing colors:-

→ contrast:- colors should be easy to distinguish

→ Accessibility:- use color-blind-friendly colors.

Effects of poor color choice:-

→ Confusing information

→ Difficult to read

→ Incorrect interpretation

→ Hides important details